

Submission Summary

Conference Name

11th International Conference on Computer Vision & Image Processing (CVIP 2026)

Track Name

CVIP2026-Phase-II

Paper ID

572

Paper Title

Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement

Abstract

Magnetic Resonance Imaging (MRI), Computed Tomography (CT) and ultrasound are commonly used medical imaging techniques for diagnosis and treatment planning. But images acquired from such systems are commonly contaminated with noise during acquisition, transmission or reconstruction. Denoising is therefore crucial for improving image quality without compromising critical anatomical features and diagnostic information. This study presents a detailed comparative analysis of various denoising techniques such as classical spatial filters, edge preserving methods and transform domain approaches. This includes Mean, Median, Gaussian, Wiener, Bilateral, and Anisotropic Diffusion filters, as well as advanced approaches such as Non-Local Means, Wavelet Transform, Curvelet Transform, Moving Average, Exponential Moving Average, and a hybrid Median-Mean filtering strategy. The performance of these techniques was quantitatively evaluated using image quality metrics, including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR). Experimental results demonstrate that transform domain methods and Non-Local Means filtering achieve superior structural preservation and higher perceptual quality, while classical filters provide lower computational complexity and faster processing. The findings of this study offer valuable insights for selecting suitable denoising techniques for different medical imaging modalities and clinical applications, contributing to improved reliability in medical image analysis and computer-aided diagnosis systems.

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Primary Subject Area

Medical Image Analysis

Submission Files

Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement.pdf (1 Mb, 2026-05-02 15:53:41 GMT+5:30)



1 of 7,388

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Hello,

The following submission has been created.

Track Name: CVIP2026-Phase-II

Paper ID: 572

Paper Title: Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement

Abstract:
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Primary Subject Area: Medical Image Analysis

Secondary Subject Areas: Not Entered

Submission Files:
Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement.pdf (981 Kb, Sat, 02 May 2026 10:23:41 GMT)

Submission Questions Response: Not Entered

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